POWER INPUT 24 V
E-STOP + WATCHDOG OUTPUT RAILS
RASPBERRY PI 5 HEADER
OUTPUT EXPANDER + RELAY DRIVER
RELAY K1
RELAY K2
RELAY K3
RELAY K4
RELAY K5
RELAY K6
RELAY K7
RELAY K8
SOLENOID DRIVERS (valves, cylinders)
INPUT EXPANDER
ISOLATED INPUTS 1–4
ISOLATED INPUTS 5–8
ISOLATED INPUTS 9–12
ISOLATED INPUTS 13–16
GAUGE / ANALOG INPUTS 0–10 V
RS-485 (turbo pump controllers, multi-drop)
RS-232
MECHANICAL

DESIGN NOTES
1. Output power: +24V -> K9 (closed only while the E-stop loop J2 is closed) -> +24V_ESTOP -> K10 (closed only while the Pi toggles GPIO19 at >= 500 Hz; charge pump blocks DC so a hung or rebooting Pi drops it in ~100 ms) -> +24V_SAFE.
2. Solenoid valves/cylinders are powered from +24V_SAFE only. Relay coils select their rail with JP1–JP8: default K1 (mechanical pump) on +24V_ESTOP so the backing pump keeps running if the Pi hangs; K2–K8 on +24V_SAFE.
3. All MCP23017 outputs are held in reset by R10 until the Pi drives GPIO22 high; every driver input has a pull-down.
4. The Pi 5 V pins are NOT connected. Board logic runs from the Pi 3V3 pin (< 60 mA). Power the Pi from its own USB-C supply.
5. K9 is a standard relay, not a safety relay. For a category-rated E-stop, wire the safety relay output into J2.
6. Relay contacts are for permissive/remote inputs of the tool's own pump controllers – not for pump mains power.
7. Inputs: 24 V sourcing sensors or dry contacts between +24V_AUX and DIn; DI_COM of each group to 0 V of that supply.
8. Analog: 0–10.5 V into 90k; ADC sees Vin x 0.2444 (PGA +/-4.096 V). Calibrate per gauge in software.
9. I2C: U1 outputs 0x20, U3 inputs 0x21, U9 ADC 0x48, U10 ADC 0x49 (Pi I2C1, GPIO2/3).
10. UART0 GPIO14/15 = RS-485 (DE = GPIO18). UART4 GPIO12/13 = RS-232 (dtoverlay=uart4-pi5).
11. Channel map = glasgow_service vacuumSystem.json / SBC Vacuum Controller dashboard:
    A0 K1 MechanicalVacuumPump (ready B0=DI1, fault DI5, gauge AI1)
    A1 K2 TurboVacuumPump (at speed B1=DI2, fault DI6, gauge AI2, speed AI5)
    A2 K3 UHVacuumPump_1 turbo (B2=DI3, fault DI7, gauge AI3, speed AI6)
    A3 K4 UHVacuumPump_2 turbo (B3=DI4, fault DI8, gauge AI4, speed AI7)
    A4 K5 HighVoltageTransformer permissive. Wiring: WIRING.md

Outputs are OFF whenever the Pi is absent, booting, or stops its heartbeat
Generated from tools/design.py – label-connected nets
Ionbeam
Sheet: /
File: RPi5VacuumIO.kicad_sch
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